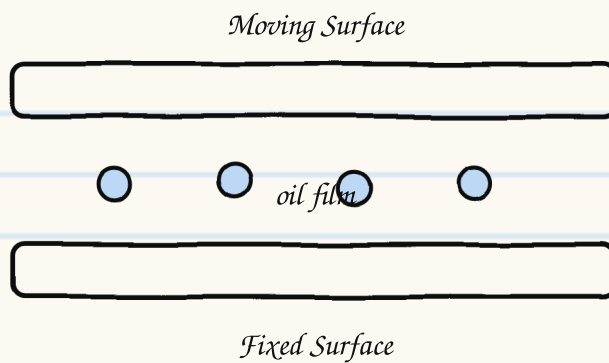


Engineering Chemistry

BT-101 | Unit 3

Lubricants & Lubrication



Topics Covered:

- *Lubricants and Lubrication - Definition*
- *Mechanism of Lubrication*
- *Classification of Lubricants*
- *Viscosity and Viscosity Index*
- *Short Notes: Flash & Fire Point, Cloud & Pour Point*

1. Lubricants and Lubrication

Lubricant: A substance (usually oil, grease or solid powder) which, when applied between two moving/sliding surfaces, **reduces friction, wear and heat generation** between them.

Lubrication: The technique/process of applying a lubricant between two surfaces in relative motion so that direct solid-to-solid contact is reduced or avoided.

Functions (Purpose) of a Lubricant

- **Reduces friction** between moving parts - saves power/energy
- **Reduces wear and tear** of machine parts - increases machine life
- **Cooling** - carries away heat generated due to friction
- **Prevents corrosion** - forms a protective film over metal surface
- Acts as a **sealant** - prevents entry of dust, dirt and moisture
- **Cleaning action** - carries away wear debris/metal particles
- Absorbs shock in gears and bearings, gives smooth running

Trick: "FRCCS" - Friction reduce, Reduce wear, Cooling, Corrosion prevent, Sealing/Shock absorb.

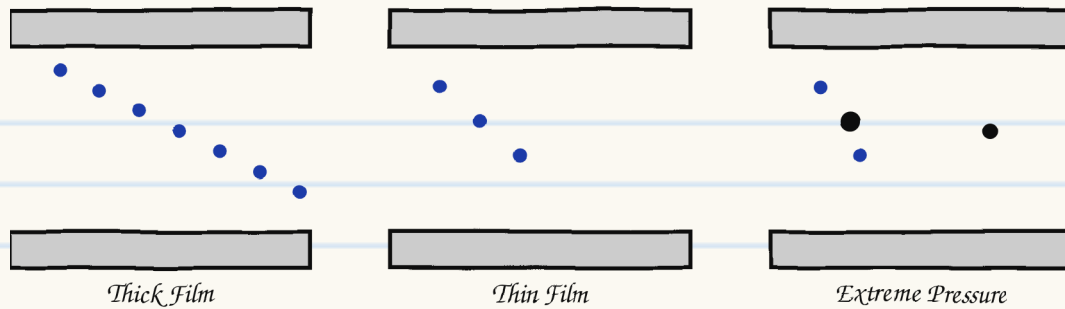
Exam Tip: "Define lubricant and lubrication. Explain functions of a lubricant." - very common short-answer question (2-5 marks, almost every year).

Quick Revision: Lubricant = substance reducing friction/wear between moving surfaces.

Lubrication = process of applying it. Main functions: reduce friction & wear, cool, protect from corrosion, seal, clean, absorb shock.

2. Mechanism of Lubrication

Depending on the load, speed and thickness of the lubricant film formed between two surfaces, lubrication mechanism is of **three types**:



(a) Thick Film / Hydrodynamic Lubrication

- A **thick continuous film** of lubricant (thickness $> 1000 \text{ \AA}$, i.e. $> 0.1 \text{ micron}$) completely separates the two moving surfaces - **no metal-to-metal contact**
- Occurs at **high speed, moderate/low load** (e.g. journal bearings running at speed)
- **Viscosity** of the lubricant is the deciding property here
- Follows laws of fluid mechanics (Newton's law of viscous flow)

(b) Thin Film / Boundary Lubrication

- Lubricant film is very **thin** ($\leq 1000 \text{ \AA}$) - surfaces are not fully separated
- **Partial metal-to-metal contact** occurs at high points (asperities) of the surface
- Occurs at **low speed, high load**, or during starting/stopping of machine
- Here '**oiliness**' (ability to stick/adhere to metal surface) of lubricant matters, not viscosity

(c) Extreme Pressure Lubrication

- Occurs under **very high load, high temperature and high speed** (e.g. hypoid gears) where even boundary lubrication film breaks down
- Ordinary lubricant film cannot withstand such severe conditions
- **EP additives** (compounds of Chlorine, Sulphur, Phosphorus) are added - they react chemically with the metal surface at high local temperature to form a protective metal-chloride/sulphide film
- This chemical film prevents seizure and welding of surfaces

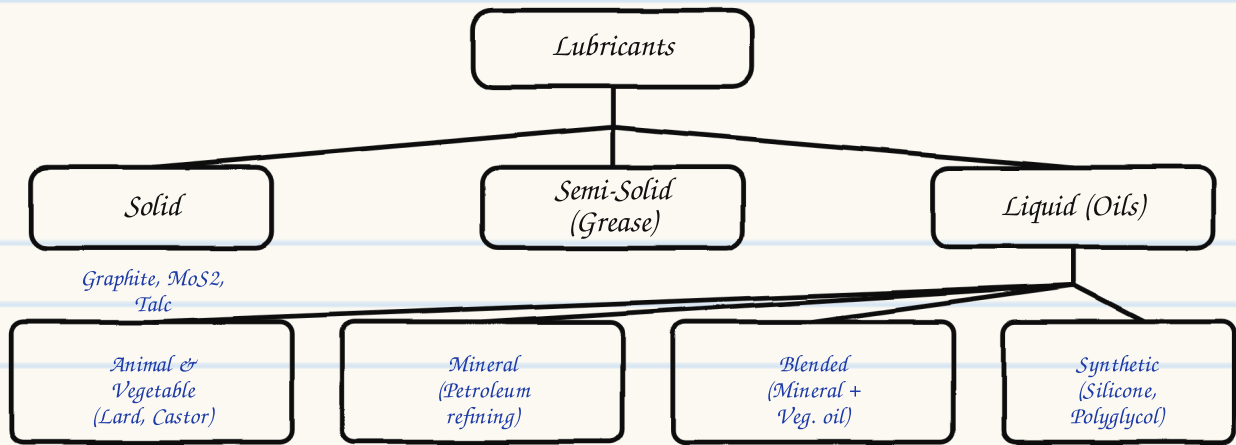
Trick: "Thick=Speed+Viscosity, Thin=Load+Oiliness, EP=Extreme+additive(Cl,S,P)" - remember by the deciding factor of each type.

PYQ Alert: "Explain mechanism of lubrication" / "Differentiate thick film, thin film and extreme pressure lubrication" - asked very frequently (5-7 marks).

Quick Revision: Thick film (hydrodynamic) - full separation, viscosity matters, high speed. Thin film (boundary) - partial contact, oiliness matters, high load/low speed. Extreme pressure - EP additives (Cl,S,P) form protective film at very high load/temp.

3. Classification of Lubricants

Lubricants are broadly classified into **three groups** based on physical state:



Liquid oil sub-types:

(a) Solid Lubricants

- Used where liquid lubricants cannot be used - i.e. at **very high temperature/load**, or where oil/grease would attract dust
- Applied as dry powder, dispersion in oil, or as a bonded film
- Examples: **Graphite, Molybdenum disulphide (MoS₂), Talc, Mica**
- Graphite & MoS₂ have a **layered (sheet-like) structure** - layers slide over each other easily, giving low friction

(b) Semi-Solid Lubricants (Greases)

- Grease = **Mineral oil + Thickening agent (metallic soap)**, e.g. Calcium/Sodium/Lithium soap
- Used where lubricant must **stay in place** and not leak/drip out - open bearings, exposed gears, wire ropes
- Works well at low speed, heavy load and where frequent lubrication is difficult

(c) Liquid Lubricants (Lubricating Oils)

Most widely used type of lubricant. Further sub-divided into 4 categories:

Type	Source / Example	Remark
Animal & Vegetable oils	Lard oil, Castor oil, Rapeseed oil	Good oiliness but poor thermal/oxidation stability; costly - rarely used alone now
Mineral oils	Obtained from petroleum refining (distillation)	Cheap, most widely used; poor natural oiliness, needs additives
Blended oils	Mineral oil + small % of vegetable/animal oil	Combines low cost of mineral oil with good oiliness of vegetable oil
Synthetic oils	Silicone oils, Polyglycols, Esters	Used for extreme temperature conditions (very high or very low temp), aircraft/space applications

Trick: "SSL - Solid, Semi-solid (grease), Liquid" and for liquid: "AMBS" - Animal & Veg, Mineral, Blended, Synthetic.

PYQ Alert: "Classify lubricants with examples" - very commonly asked (3-5 marks); sometimes asked as a diagram/flowchart question.

Quick Revision: Lubricants: Solid (graphite, MoS₂) - high temp/load; Semi-solid/Grease (oil+soap) - stays in place; Liquid oils - Animal/Veg, Mineral, Blended, Synthetic (most common, mineral oil cheapest & most used).

4. Viscosity and Viscosity Index

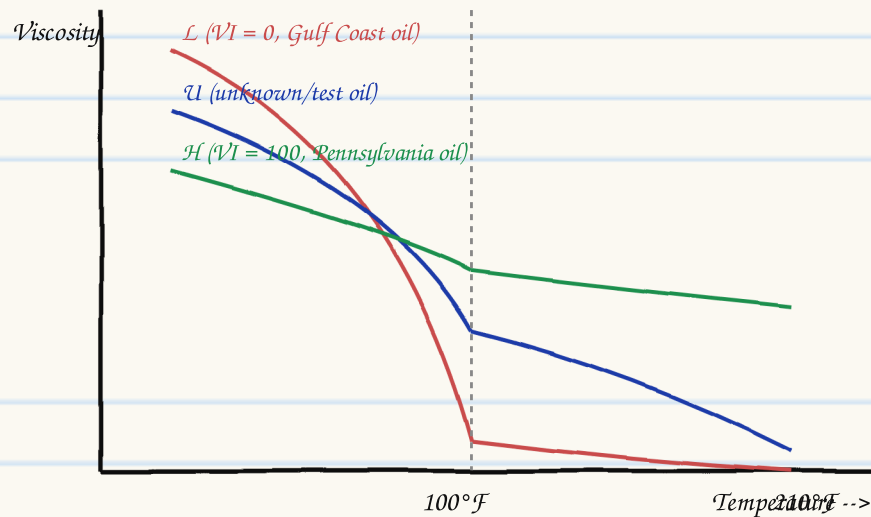
Viscosity: It is the *internal resistance offered by a fluid to its own flow*. It is a measure of the "thickness" or flow-resistance of a lubricating oil.

- Higher viscosity $\Rightarrow$ oil flows slowly (thick); Lower viscosity $\Rightarrow$ oil flows easily (thin)
- Units: **Poise** (CGS system), **Pascal-second, Pa.s** (SI system)
- **Important:** Viscosity of a liquid **decreases as temperature increases** (opposite behaviour to gases)
- Viscosity decides how well the lubricant film can be maintained between two surfaces under load - it is the key property in selecting a lubricating oil

Viscosity Index (VI)

Viscosity Index: It is an *empirical number* that indicates how much the viscosity of an oil changes with change in temperature. A good lubricant should have **minimum change** in viscosity over a wide temperature range.

- **High VI oil** $\Rightarrow$ viscosity changes very little with temperature (desirable - "all season" oil, good for automobile engines where temperature varies from cold start to hot running)
- **Low VI oil** $\Rightarrow$ viscosity changes a lot with temperature (undesirable)



Dean and Davis Scale (measurement basis)

Two reference oils are chosen, both having the **same viscosity at 210°F** as the unknown (test) oil, but different viscosity at 100°F:

- **Pennsylvania (Gulf) oil** - shows least change in viscosity with temperature $\Rightarrow$ assigned **VI = 100**
- **Gulf Coast oil** - shows most change in viscosity with temperature $\Rightarrow$ assigned **VI = 0**

$$VI = \frac{(L - U)}{(L - H)} \times 100$$

L = viscosity of 0-VI standard oil at 100°F U = viscosity of unknown (test) oil at 100°F H = viscosity of 100-VI standard oil at 100°F (all three have equal viscosity at 210°F)

Trick: "LUH - Low, Unknown, High" in that order in the formula: $(L-U)/(L-H) \times 100$. Also remember: "High VI = High quality" (less variation = better oil).

PYQ Alert: "Define viscosity and viscosity index. Give formula and its significance" - asked very frequently (numerical based on VI formula also asked sometimes).

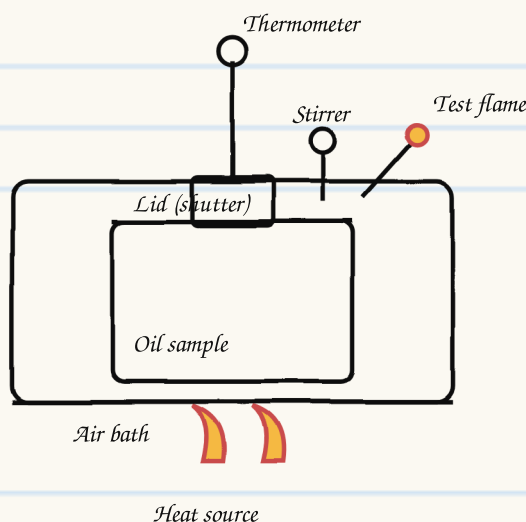
Quick Revision: Viscosity = resistance to flow, decreases with rise in temperature. VI = measure of variation of viscosity with temperature; High VI = stable oil (good). $VI = \frac{(L-U)}{(L-H)} \times 100$ (Pennsylvania=100, Gulf Coast=0).

5. Short Notes

(A) Flash Point and Fire Point

Flash Point: The **lowest temperature** at which an oil gives off enough vapour that it **momentarily catches fire (flashes)** when a small test flame is brought near its surface, but does not continue to burn.

Fire Point: The **lowest temperature** at which the vapour of the oil, once ignited, **continues to burn for at least 5 seconds** continuously. Fire point is always a **few degrees higher** than flash point.



Apparatus & Method

- Determined using **Pensky-Martens (closed cup) apparatus** or **Cleveland (open cup) apparatus**
- Oil is heated slowly and steadily with continuous stirring; a small test flame is applied at regular temperature intervals through the shutter
- Temperature at first flash = Flash point; temperature at which burning continues (≥ 5 sec) = Fire point

Significance

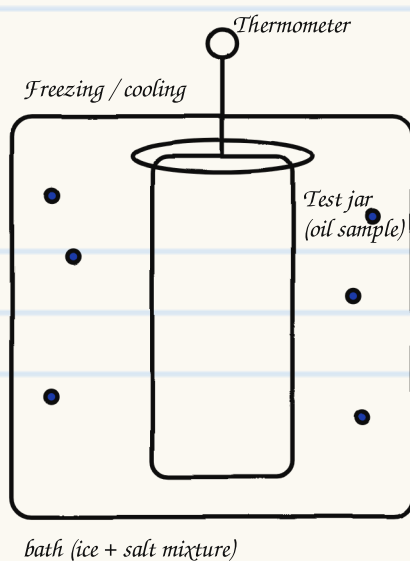
- Indicates the **fire/explosion hazard** and safe handling/storage temperature of the lubricant
- A good lubricant (especially for use at high operating temperatures, e.g. IC engines) should have a **high flash and fire point** for safety

Trick: "Flash comes first (momentary spark), Fire stays longer (continuous burn, higher temp)."

(B) Cloud Point and Pour Point

Cloud Point: The **temperature at which an oil (containing dissolved wax) begins to turn cloudy or hazy** in appearance when it is cooled slowly, due to the wax starting to separate out as tiny crystals.

Pour Point: The **lowest temperature** at which the oil **just ceases to flow / pour** under specified test conditions when cooled further - the wax crystals form a rigid network that traps the oil.



Method

- A sample of oil is taken in a test jar fitted with a thermometer
- The jar is placed in a cooling/freezing bath (ice + salt mixture) and cooled slowly
- Jar is periodically tilted to check flow - temperature of first haziness = Cloud point; temperature at which oil does not flow even after tilting for 5 seconds = Pour point

Significance

- Very important for lubricants meant to be used in **cold climates / winter conditions**
- A good lubricant for cold conditions should have a **low pour point**, so that it remains fluid and can lubricate the engine properly even during a cold start

Trick: "Cloud comes before Pour" - oil first turns cloudy (Cloud point, higher temp), then stops flowing on further cooling (Pour point, lower temp).

PYQ Alert: "Write short note on Flash & Fire point" and "Write short note on Cloud & Pour point" - both asked almost every year as 2-3 mark short questions.

Quick Revision: Flash point (momentary flash) < Fire point (burns ≥ 5 sec) - high value desirable for safety. Cloud point (wax starts appearing, oil turns hazy) > Pour point (oil stops flowing) - low value desirable for cold climates.

Unit 3 - Final Quick Revision Sheet

Lubricant & Lubrication

- Lubricant reduces friction, wear, heat between moving surfaces
- Functions: friction reduce, cool, protect, seal, clean, absorb shock

Mechanism

- Thick film (hydrodynamic) - full separation, viscosity matters
- Thin film (boundary) - partial contact, oiliness matters
- Extreme pressure - EP additives (Cl, S, P) form protective film

Classification

- Solid (graphite, MoS₂), Semi-solid (grease), Liquid (oils)
- Liquid: Animal/Veg, Mineral, Blended, Synthetic

Viscosity & VI

- Viscosity = resistance to flow; decreases with rise in temp
- $VI = (L-U)/(L-H) \times 100$; High VI = stable, good oil

Flash / Fire Point

- Flash = momentary ignition temp; Fire = continuous burning ($\geq 5s$) temp
- High value desirable for safety

Cloud / Pour Point

- Cloud = wax starts appearing (haziness)
- Pour = lowest temp oil still flows; low value good for winter

Most Important (High-weightage) Questions:

1. Define lubricant & lubrication. Explain its functions. (Almost every year)
2. Explain mechanism of lubrication - thick film, thin film, extreme pressure. (Very frequently asked)
3. Classify lubricants with examples (flowchart/diagram). (2019,21,23)
4. Define viscosity and viscosity index. Derive/state VI formula. (Frequently asked, sometimes numerical)
5. Short note on Flash & Fire point / Cloud & Pour point. (Every year, 2-3 marks each)

One-line memory chain: "Lubricant reduces friction → works in 3 regimes (thick/thin/EP) → comes as solid/semi-solid/liquid → liquid oils judged by Viscosity & VI → safety by Flash/Fire point → cold performance by Cloud/Pour point."